

Mathematical Foundations of the Qwen3-VL Taste-Annotation Evaluation Suite

Abstract

This note formalizes the statistics used throughout the five-benchmark evaluation of Qwen3-VL-30B as an automated taste-annotation labeler (`qwen_evaluation_suite/`). It covers: (1) pairwise Cohen’s κ and the “agreement envelope” framing used to compare model–human agreement against human–human agreement; (2) plurality consensus labeling and the confusion-matrix decomposition used under structurally missing (conditional) fields; (3) the cosine-similarity relaxation of exact-match accuracy used for the free-text `noun` field, with a worked decomposition matching the observed numbers; (4) classical test theory applied to rater reliability, proving that self-consistency (test–retest agreement) is necessary but not sufficient for validity, with the suite’s own “self-fed” bias as the canonical counterexample; and (5) a significance test formalizing the claim that five independent ablations all failing to move human-fed agency accuracy off 0% constitutes strong evidence of a genuine perceptual ceiling rather than a fixable prompting gap.

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1 Pairwise Cohen’s κ and the agreement envelope

1.1 Setup

Let a *rater* be either a human annotator or the model, and let two raters A, B each assign one label from a finite category set $\mathcal{C} = \{c_1, \dots, c_K\}$ (e.g. $\mathcal{C} = \{\text{eating, possibly-eating, not-eating}\}$ for the `scenario` field) to each of n clips they both labeled. Write $y_i^A, y_i^B \in \mathcal{C}$ for their labels on clip i .

Definition 1 (Observed and chance agreement). *The observed agreement is*

$$p_o(A, B) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[y_i^A = y_i^B].$$

Let $\pi_k^A = \frac{1}{n} \sum_i \mathbb{I}[y_i^A = c_k]$ and $\pi_k^B = \frac{1}{n} \sum_i \mathbb{I}[y_i^B = c_k]$ be each rater’s marginal frequency for category c_k . The chance agreement under the assumption that A and B label independently with these marginals is

$$p_e(A, B) = \sum_{k=1}^K \pi_k^A \pi_k^B.$$

Definition 2 (Cohen’s κ).

$$\kappa(A, B) = \frac{p_o(A, B) - p_e(A, B)}{1 - p_e(A, B)}.$$

$\kappa = 1$ means perfect agreement beyond chance; $\kappa = 0$ means the raters agree exactly as often as two independent labelers with the same marginals would; $\kappa < 0$ means they agree *less* than chance would predict (systematic disagreement).

1.2 Pooled multi-rater estimate

The suite has 11 human raters $H_1, \dots, H_{11}$ (each labeling a subset of the 506-clip pool, up to 3 raters per clip) plus the model, Q. Rather than Fleiss’ κ (which requires every clip to have the same number of raters), Benchmark 3 uses a *pooled pairwise* estimator: for a field f , collect every $(y_j^{H_i}, y_j^{H_k})$ pair for every clip j that both H_i and H_k labeled, over all $\binom{11}{2}$ rater pairs, into one pooled sample $\mathcal{P}_{\text{hh}}$; separately collect every $(y_j^{H_i}, y_j^Q)$ pair into $\mathcal{P}_{\text{qh}}$. Then

$$\kappa_{\text{hh}} = \kappa(\mathcal{P}_{\text{hh}}), \quad \kappa_{\text{qh}} = \kappa(\mathcal{P}_{\text{qh}}),$$

using the same observed/chance-agreement formulas with $\mathcal{P}$ in place of a single rater pair’s clip set.

Remark 1 (Why pooling is a reasonable but conservative estimator). *Pairs drawn from the same clip (e.g. (H_i, H_k) and (H_i, H_l) on clip j) are not independent observations, since they share $y_j^{H_i}$. This does not bias $\hat{\kappa}$ but does mean its naive standard error is an underestimate; the suite reports point estimates only; a clip-clustered bootstrap would be the appropriate correction if confidence intervals were needed.*

1.3 The agreement-envelope test

The scientific question Benchmark 3 asks is not “is Qwen accurate?” in isolation, but:

Is κ_{qh} (and $p_{o,\text{qh}}$) inside the range humans achieve with each other, κ_{hh} ?

This reframes evaluation around a data-dependent ceiling rather than a fixed target of 100%, because for genuinely ambiguous clips even $\kappa_{\text{hh}} < 1$ is the best achievable score — no rater, human or model, can exceed the label’s intrinsic annotator disagreement. Formally, if we posit a noisy-annotator model where each human rater observes the true (possibly ill-defined) label Y_i through a rater-specific noise channel, κ_{hh} estimates the *signal-to-noise ratio of the labeling task itself*, and a rater (model or human) with $\kappa_{\text{model},h} \approx \kappa_{\text{hh}}$ is, by this criterion, indistinguishable from another human annotator.

Field	κ_{hh}	κ_{qh}	$p_{o,\text{hh}}$	$p_{o,\text{qh}}$
scenario	0.44	0.42	64.8%	64.3%
verb	0.62	0.26	69.5%	40.5%
condition	0.28	0.28	71.5%	76.4%
flavor	0.52	0.26	64.6%	44.2%
noun	0.55	0.48	60.7%	57.8%

Table 1: Pooled-pairwise agreement, Benchmark 3 (full 506-clip pool).

By this test, **scenario**, **condition**, and **noun** are inside the human envelope ($\kappa_{\text{qh}} \approx \kappa_{\text{hh}}$ or, for **condition**, exactly equal), while **verb** and **flavor** fall well outside it (κ_{qh} roughly 0.4× the human value in both cases) — the same conclusion drawn qualitatively from the rater-agreement heatmap, now with a single scalar per field.

2 Plurality consensus and confusion under structural missingness

2.1 Plurality (majority-vote) consensus labels

For a clip j labeled by raters $H_{i_1}, \dots, H_{i_m}$ ($m \geq 2$), let $v_j(c) = |\{r : y_j^{H_{i_r}} = c\}|$ count votes for category c .

Definition 3 (Plurality consensus). *Let $c_{(1)}, c_{(2)}$ be the categories with the largest and second-largest vote counts, $v_{(1)} \geq v_{(2)}$ (ties broken arbitrarily among tied runners-up). The consensus label is defined only when the winner is strict:*

$$\hat{Y}_j = \begin{cases} c_{(1)} & \text{if } v_{(1)} > v_{(2)}, \\ \text{undefined} & \text{if } v_{(1)} = v_{(2)} \text{ (exact tie)}. \end{cases}$$

With $m = 3$ raters, a tie is possible only when all three raters disagree ($v_{(1)} = v_{(2)} = v_{(3)} = 1$); a 2-1 split always yields a defined consensus. With $m = 2$, consensus is defined exactly when the two raters agree ($v_{(1)} = 2$). Benchmark 3’s accuracy-vs-consensus metric,

$$\text{Acc}_{\text{cons}}(f) = \frac{1}{|\mathcal{J}_f|} \sum_{j \in \mathcal{J}_f} \mathbb{1}[y_j^{\text{Q}} = \hat{Y}_j], \quad \mathcal{J}_f = \{j : \hat{Y}_j \text{ defined for field } f\},$$

is therefore computed over clips where the humans reached a strict plurality, which structurally excludes the hardest 3-way-split clips and should be read as an upper bound on how well Qwen matches human judgment on the pool as a whole.

2.2 Difficulty transfer as a stratified accuracy

Partition $\mathcal{J}_f$ by the winning margin $v_{(1)} - v_{(2)}$ (for $m = 3$: *unanimous* when $v_{(1)} = 3$, *majority* when $v_{(1)} = 2$; for $m = 2$: *agree* when $v_{(1)} = 2$, and treat a raw 1-1 *split* as its own stratum where a prediction counts as correct if it matches *either* rater). The *difficulty-transfer* statistic is the plurality accuracy computed separately within each stratum,

$$\text{Acc}_{\text{cons}}(f \mid \text{stratum } s) = \frac{1}{|\mathcal{J}_f \cap s|} \sum_{j \in \mathcal{J}_f \cap s} \mathbb{1}[y_j^{\text{Q}} = \hat{Y}_j].$$

Proposition 1 (Monotone difficulty transfer implies shared ambiguity). *Suppose model errors are conditionally independent of human disagreement given the (unobserved) intrinsic ambiguity $A_j \in [0, 1]$ of clip j , and that A_j jointly increases both the probability a human rater deviates from the modal label and the probability the model deviates from it. Then $\text{Acc}_{\text{cons}}(f \mid s)$ is monotonically decreasing as s ranges from unanimous to split. Observing this monotone decrease (empirically: 84.6% $\rightarrow$ 42.6% for scenario, unanimous vs. majority) is evidence for the shared-ambiguity model over the alternative hypothesis that Qwen’s errors are i.i.d. and uncorrelated with clip difficulty, since the latter predicts a flat $\text{Acc}_{\text{cons}}(f \mid s)$ across strata.*

2.3 Structural (MNAR) missingness of conditional fields

The four fields `verb`, `additional_condition`, `noun`, `flavor_profile` are only elicited from a rater when that rater’s own `scenario` answer is not `not-eating`. This is missingness *not at random* (MNAR) in the classical taxonomy: whether $y_{j,f}^r$ is observed for rater r , field f , clip j is a deterministic function of that same rater’s `scenario` label, $R_j^r = \mathbb{1}[y_{j,\text{scenario}}^r \neq \text{not-eating}]$, which is exactly the mechanism that generates the field values themselves.

Definition 4 (Comparable-pair restriction). *For a conditional field f and a pair of raters (A, B) on clip j , the pair contributes to the pooled comparison only when both raters observed the field:*

$$(y_{j,f}^A, y_{j,f}^B) \in \mathcal{P}_f \iff R_j^A = 1 \text{ and } R_j^B = 1.$$

Remark 2 (Why this restriction is necessary, and what it costs). *Comparing on $R_j^A = R_j^B = 1$ only is required because a blank value means “not elicited,” not “elicited and equal to the empty string”—treating it as its own category would inflate agreement whenever two raters agree on `scenario` being `not-eating` (the fields are trivially equal by construction, contributing no information about the conditional field itself). The cost is a selection effect: $\mathcal{P}_f$ is implicitly conditioned on scenario agreement between A and B , so κ_{qh} for `verb/flavor/noun/condition` already excludes the subset of clips where Qwen and the human disagree about whether eating is happening at all — the reported n for these fields ($n=432$ for Qwen $\leftrightarrow$ human vs. $n=347$ for human $\leftrightarrow$ human, Benchmark 3) reflects exactly this conditioning, not a data-loss artifact.*

3 Cosine similarity as a soft relaxation of exact-match accuracy

3.1 Embedding cosine similarity

The `noun` field is free text (“cereal”, “sippy cup”, “baked beans”, ...), so exact-string accuracy penalizes correct-but- differently-worded answers as harshly as wrong ones. Let $e : \mathcal{W} \rightarrow \mathbb{R}^d$

be a sentence-embedding map (MiniLM, $d = 384$) with $\|e(w)\|_2 = 1$ for all w (embeddings are L2-normalized before comparison). For ground-truth noun g_i and predicted noun $\hat{g}_i$ on clip i ,

$$\text{sim}(g_i, \hat{g}_i) = \cos \theta_i = \frac{e(g_i) \cdot e(\hat{g}_i)}{\|e(g_i)\| \|e(\hat{g}_i)\|} = e(g_i) \cdot e(\hat{g}_i) \in [-1, 1],$$

the last equality using unit normalization. Because e is trained so that semantically related words are embedded nearby, sim acts as a continuous relaxation of the indicator $\mathbb{I}[g_i = \hat{g}_i]$: identical strings give $\text{sim} = 1$ exactly, while near-synonyms (“sippy cup” vs. “bottle”) give intermediate values and unrelated words give values near 0 or below.

3.2 Soft accuracy and its relation to exact-match accuracy

Definition 5 (Soft accuracy at threshold τ).

$$\text{Acc}_{\text{soft}}(\tau) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sim}(g_i, \hat{g}_i) \geq \tau].$$

Exact-match accuracy is the degenerate case $\text{Acc}_{\text{exact}} = \text{Acc}_{\text{soft}}(1)$ under an embedding where distinct strings almost never map to identical vectors, and the mean similarity $\overline{\text{sim}} = \frac{1}{n} \sum_i \text{sim}(g_i, \hat{g}_i)$ reported throughout the suite is not a threshold accuracy but a continuous summary of the same quantity.

Proposition 2 (Decomposition of mean similarity by exact-match status). *Let $\mathcal{M} = \{i : g_i = \hat{g}_i\}$ (exact matches, contributing $\text{sim}_i = 1$) and $\mathcal{M}^c$ its complement, with $m = |\mathcal{M}^c|/n$ the mean similarity restricted to mismatches. Then*

$$\overline{\text{sim}} = \text{Acc}_{\text{exact}} \cdot 1 + (1 - \text{Acc}_{\text{exact}}) \cdot \bar{m}, \quad \bar{m} = \frac{\overline{\text{sim}} - \text{Acc}_{\text{exact}}}{1 - \text{Acc}_{\text{exact}}}.$$

Proof. Split the sum defining $\overline{\text{sim}}$ over $\mathcal{M}$ and $\mathcal{M}^c$; every term in $\mathcal{M}$ equals 1 by definition of exact match, and $|\mathcal{M}|/n = \text{Acc}_{\text{exact}}$. Rearranging gives $\bar{m}$. $\square$

3.3 Worked example (Benchmark 1)

Benchmark 1 reports, on the annotator-5 eval set restricted to human eating/possibly-eating clips, $\text{Acc}_{\text{exact}} = 70.8\%$ ($n = 48$) and $\overline{\text{sim}} = 82.0\%$ ($n = 46$; two additional clips were excluded where the human noun was blank, so the identity below is approximate rather than exact over an identical index set). Applying Proposition 2 with these figures,

$$\bar{m} \approx \frac{0.820 - 0.708}{1 - 0.708} \approx 0.384,$$

i.e. on the $\approx 29\%$ of clips where Qwen’s noun did *not* exactly match the human’s, the two words were on average only weakly related (cosine ≈ 0.38) rather than close synonyms. Cross-checking against the raw pairs reported in the suite (e.g. **cereal** $\rightarrow$ **(none)** at 0.13, **banana** $\rightarrow$ **bottle** at 0.42, **beans** $\rightarrow$ **baked beans** at 0.81) confirms the mismatches are a mix of outright misses (Qwen abstaining or picking an unrelated object) and near-misses (a related but non-identical object), consistent with $\bar{m}$ sitting well below 1 but above 0.

Remark 3 (Why $\overline{\text{sim}} \geq \text{Acc}_{\text{exact}}$ is not automatic). *Proposition 2 shows $\overline{\text{sim}} > \text{Acc}_{\text{exact}}$ exactly when $\bar{m} > 0$, i.e. when the model’s mistakes are, on average, still semantically closer to the truth than two independently drawn nouns would be. This is an empirical property of the model’s error distribution, not a mathematical guarantee — a model that fails by naming a semantically unrelated object every time it is wrong would have $\bar{m} \leq 0$ and could show $\overline{\text{sim}} < \text{Acc}_{\text{exact}}$.*

4 Classical test theory: reliability is not validity

4.1 Reliability as test–retest agreement

Classical test theory models an observed measurement as $X = T + E$, a true-score component T plus measurement error E with $E[E] = 0$ and $\text{Cov}(T, E) = 0$. The *reliability* of the measurement procedure is

$$\rho_{XX'} = \frac{\text{Var}(T)}{\text{Var}(X)} = \frac{\text{Var}(T)}{\text{Var}(T) + \text{Var}(E)},$$

estimated in practice by the correlation (for continuous scores) or percent agreement / κ (for categorical labels) between two independent applications of the same measurement procedure to the same items — exactly Benchmark 5’s construction: Qwen labels the same 101 clips three times with sampling ($T=0.7$), and the pooled pairwise agreement across the $\binom{3}{2} = 3$ run-pairs estimates $\rho_{XX'}$ for each field.

Field	Qwen self-agreement $\hat{\rho}_{XX'}$	human $\leftrightarrow$ human agreement
scenario	88.1%	64.8%
verb	91.2%	69.5%
condition	90.4%	71.5%
flavor	93.4%	64.6%
noun	87.5%	60.7%

Table 2: Benchmark 5, Part A: Qwen is more self-consistent than independent humans are with each other, on every field.

4.2 The attenuation bound: reliability limits, but does not imply, validity

Let Y denote the (human-consensus) ground truth and X the model’s label, both modeled as noisy measurements of a shared underlying construct. The classical *attenuation formula* relates the correlation between two noisy measurements to their individual reliabilities:

$$\rho_{XY} \leq \sqrt{\rho_{XX'} \rho_{YY'}}.$$

Proposition 3 (Reliability is necessary, not sufficient, for validity). *High self-agreement $\rho_{XX'}$ upper-bounds, but does not lower-bound, agreement with ground truth ρ_{XY} : $\rho_{XX'} = 1$ is consistent with $\rho_{XY} = 0$.*

Proof. Let $X \equiv b$ be a *constant* predictor (formally, $X_i = b$ for every item i , independent of any input). Two independent applications of this “measurement procedure” agree on every item, so its test–retest agreement is 1 (perfect reliability: $\text{Var}(E) = 0$, so $\rho_{XX'} = 1$ trivially, since $\text{Var}(T) = \text{Var}(X) = 0$ degenerates the ratio to 1 by convention for a constant). Yet if the true label Y_i varies across items and is not itself constant, $\text{Cov}(X, Y) = 0$ identically (a constant has zero covariance with anything), so $\rho_{XY} = 0$. This satisfies the attenuation bound ($0 \leq \sqrt{1 \cdot \rho_{YY'}}$) with equality only failing to be tight, exhibiting a case of maximal reliability and zero validity. $\square$

Corollary 1 (Application to the observed self-fed prior). *Model the **verb** agency sub-field, restricted to clips where the true agency is **human-fed**, as $X_i \approx \text{self}$ for (nearly) every such clip i — an approximately constant predictor on this subpopulation, per Benchmark 2’s finding of 0% agreement with truth across five independent elicitation conditions (Table 3, reproduced from Section 5). By*

Proposition 3’s mechanism, this near-constant behavior is exactly what produces high measured self-agreement on the field overall (91.2%, Table above, pooling both agency subpopulations) while contributing systematically zero validity on the human-fed subpopulation specifically. Reliability computed on the marginal field distribution therefore masks a validity failure that is only visible once the field is stratified by true label — the same diagnostic principle underlying Proposition 1’s stratification for consensus accuracy.

4.3 Bias–variance view of the same phenomenon

Equivalently, write the per-item error as $\Delta_i = \mathbb{I}[X_i \neq Y_i]$ and decompose the population error rate on a subgroup s (e.g. true label = human-fed) as

$$E_s[\Delta] = \underbrace{(\Pr(X = \text{self} \mid s) - 0)}_{\text{systematic bias on } s} + \underbrace{(\text{noise-driven disagreement with } Y \text{ within } s)}_{\text{variance term}}.$$

A rater with zero variance (perfectly reliable, in the test–retest sense) but 100% systematic bias toward one label on a subgroup contributes $E_s[\Delta] = 1$ on exactly that subgroup regardless of how reliable it is elsewhere — reliability controls only the variance term, never the bias term, which is precisely why five independent interventions that only perturb *how the question is asked* (Section 5) cannot repair a bias-dominated failure mode: none of them supply the missing visual/attentional signal that would move $\Pr(X = \text{self} \mid \text{human-fed})$ away from its default.

5 Significance of the human-fed blind spot across independent interventions

5.1 A single-condition test

Benchmark 2 evaluated agency accuracy on the $n = 17$ clips with true label **human-fed** under several independent elicitation conditions:

Condition	Agency accuracy, true human-fed ($n = 17$)
Composite verb question	0%
Agency asked alone	0%
“Whose hand holds the food?” (perception probe)	6%
Describe-people-first scaffold	18%
“fed-by-another-person” rewording	0%

Table 3: Reproduced from Benchmark 2 / Benchmark 5.

Fix a null hypothesis H_0 : the model’s per-trial probability of correctly identifying agency on a true human-fed clip is some baseline rate p_0 representative of “the model can perceive agency at least as well as it does on the easier self-fed subgroup” (empirically $\approx 90\%$ there), or, more conservatively, $p_0 = 0.5$ (“no better than an agency coin flip”). Under H_0 , with K the number of correct calls out of $n = 17$ i.i.d. trials, $K \sim \text{Binomial}(17, p_0)$, and the observed $K = 0$ has one-sided p -value

$$\Pr(K \leq 0 \mid n = 17, p_0) = (1 - p_0)^{17}.$$

$$p_0 = 0.5 : \quad (0.5)^{17} \approx 7.6 \times 10^{-6}, \quad p_0 = 0.9 : \quad (0.1)^{17} \approx 10^{-17}.$$

Either way, observing zero correct calls out of 17 is not plausibly explained by ordinary sampling noise around a reasonable baseline rate — it is evidence the true per-trial success probability on this subgroup is near zero, not merely below its self-fed-subgroup counterpart.

5.2 Combining across independent conditions: Fisher’s method

Table 3 contains five *semi-independent* probes (independent in the sense that each perturbs a different aspect of elicitation — question framing, isolated sub-question, pure perception, reasoning scaffold, label vocabulary — rather than being repeated trials of the identical prompt). Let p_i be the one-sided binomial p-value computed as above (under $H_0: p_0 = 0.5$) for condition $i \in \{1, \dots, 5\}$, using the reported accuracy to back out an equivalent integer success count $K_i = \text{round}(0.17 \times \text{acc}_i)$ out of $n = 17$:¹

Definition 6 (Fisher’s combined probability test).

$$\chi_{\text{combined}}^2 = -2 \sum_{i=1}^5 \ln p_i \sim \chi_{2 \times 5=10}^2 \quad \text{under } H_0.$$

With $p_1 = p_2 = p_5 = \Pr(K \leq 0 \mid n=17, p_0=0.5) \approx 7.63 \times 10^{-6}$ (the three 0% conditions), $p_3 = \Pr(K \leq 1 \mid n=17, p_0=0.5) \approx 1.37 \times 10^{-4}$, and $p_4 = \Pr(K \leq 3 \mid n=17, p_0=0.5) \approx 6.36 \times 10^{-3}$,

$$\chi_{\text{combined}}^2 \approx -2(3 \ln(7.63 \times 10^{-6}) + \ln(1.37 \times 10^{-4}) + \ln(6.36 \times 10^{-3})) \approx 98.6,$$

against a χ_{10}^2 null distribution whose 99.9th percentile is about 29.6 — the combined statistic exceeds it by more than 3 $\times$, giving a combined p-value (numerically, $\text{sf}_{\chi_{10}^2}(98.6) \approx 1.0 \times 10^{-16}$) many orders of magnitude below any conventional significance threshold.

Remark 4 (What this rules in and rules out). *This computation does not claim the five conditions are statistically independent in the strict i.i.d. sense (they share the same 17 clips and the same underlying model weights), so the nominal χ_{10}^2 p-value should be read qualitatively rather than taken at face value. What it does formalize is the qualitative argument used in Section 4: a fixable prompting deficiency predicts that at least one of five substantially different framings (especially the pure-perception probe, which asks nothing about the taxonomy at all) should recover non-trivial accuracy, whereas a perceptual deficiency in how the visual encoder or attention mechanism attributes a hand to a person predicts exactly the observed pattern: uniformly near-zero accuracy regardless of how the question is phrased, with only the scaffold condition — which adds an explicit step of describing what is seen before answering — producing a partial (and still far from adequate) recovery to 18%.*

5.3 Interaction with the few-shot result

Benchmark 5, Part B adds a sixth and seventh condition (4-shot and 8-shot in-context demonstrations, the latter explicitly including two human-fed examples) and observes 0% agency accuracy on true human-fed clips at *both* shot counts. Extending the binomial argument, providing correctly labeled human-fed exemplars in-context is the single intervention most directly targeted at a demonstration-availability explanation of the failure (“the model has never been shown what human-fed looks like in this schema”); its failure to move K away from 0 is the strongest single piece of evidence in the suite that the deficit is not remediable by prompting or in-context learning

¹ $K_i = 0$ for the three 0% conditions, $K_i = 1$ for the 6% probe, $K_i = 3$ for the 18% scaffold.

at the current model scale, and would require either fine-tuning on labeled human-fed examples or an architectural/attention-level intervention (e.g. person-conditioned attention, as sketched by the scaffold condition’s partial success) to resolve.